

Pyramid Codec

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Pyramid Codec is an image compression application based on Gaussian and Laplacian image pyramids. It compresses RGB images by repeatedly smoothing and downsampling them while storing Laplacian residuals required for reconstruction. The GUI allows users to compress, save, reload, reconstruct and export images.

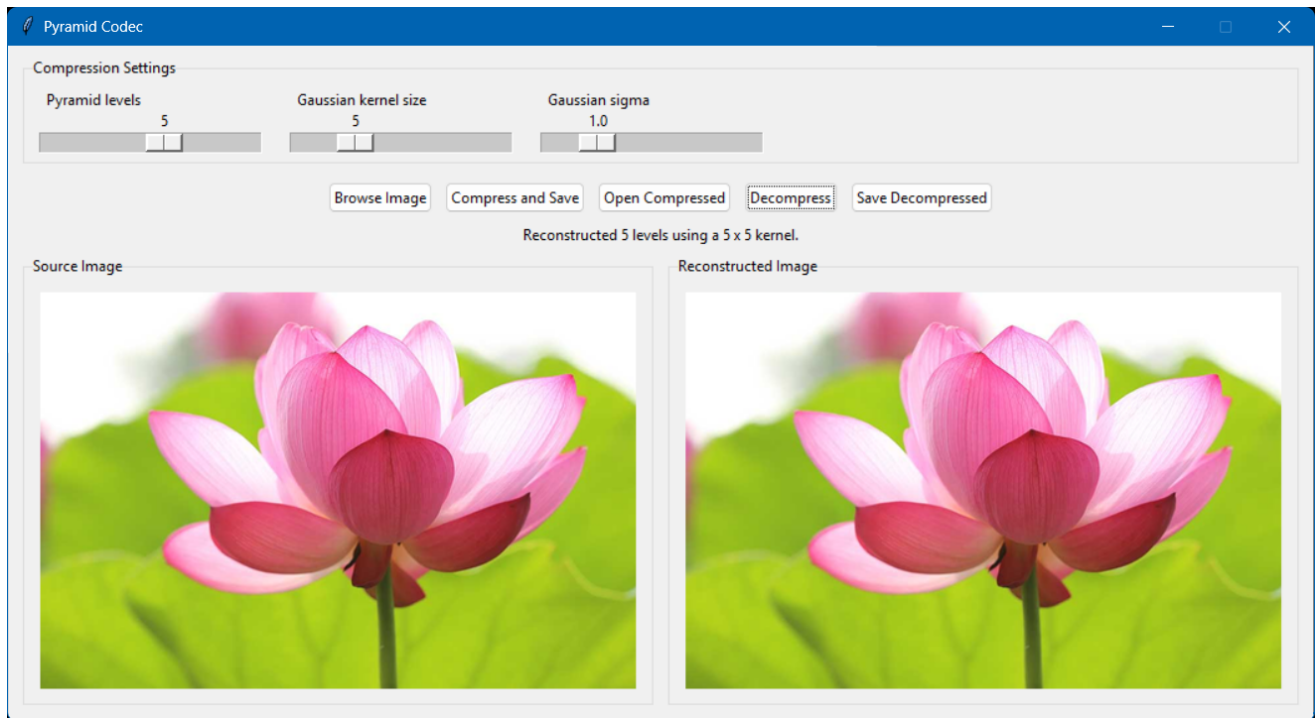
Compression Pipeline

1. Load the RGB image.
2. Blur using a Gaussian filter.
3. Downsample by a factor of two.
4. Expand the smaller image.
5. Compute and store the Laplacian residual.
6. Repeat for the selected number of pyramid levels.
7. Quantize residuals and save them together with the smallest image.

Input Image



Graphical User Interface



Reconstructed Image



Conclusion

The reconstructed image closely matches the original image, demonstrating that Laplacian pyramids preserve image detail efficiently while reducing storage requirements. This project also

provides a reusable and modular framework for experimenting with multiscale image compression techniques.